

SpotMicro

Bootloader design

⚠ Remember, the bootloader is always the entrypoint of the code

Responsibilities of the bootloader

1. Decide whether to boot or enter boot mode. Since we can only start up a "flash new code" script minutes after boot, the best option is to make the MCUs boot by default, but always listen to some `ENTER_BOOTLOADER_MODE` option. When receiving it, the MCU would:
 1. Disable motors and other peripherals safely
 2. Disable used PPL
 3. Disable interrupts and clear pending ones
 4. Store a flag in flash like `BOOTLOADER_REQUESTED = 1`
 5. Trigger a software reset

Addressing

STM32 provides a UID for each device, stored at the following memory locations: (low) `0x1FFF7590`, `0x1FFF7594`, `0x1FFF7598` (high).

The most flexible option is to make the MCUs read said UID, send it to the jetson, and receive a shorter, dynamic ID back.

Does this work in bootloader mode as well?

It doesn't make sense. The best approach is to just use the UID for the boot part. (According to cgpt) this match should only be done in application code.